GC University Lahore

Department of Computer Science

HONEY PROCESSING ENTERPRISE

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1. Project Description

The Honey Processing Enterprise wants to maintain its complete operational and management data using a Database Management System. The organization deals with multiple processes including human resource management, farm management, honey production, resource allocation, and sales distribution.

The system starts with the Human Resource (HR) department, where vacancies are created based on alerts generated by the management. Candidates apply for these vacancies, and after selection, they are assigned roles and departments, becoming employees of the organization. The system maintains employee records, their attendance, salary calculations, and transaction details.

The organization operates farms where trees, plants, and water resources are managed. Employees purchase hives from suppliers and install them on trees through the hive management team. The farm management team monitors the health of trees, plants, and water resources. Any issues or updates are reported to the manager, and reports may trigger alerts for further actions.

Once the hives are ready, they are collected and sent to the Honey Extraction Department. This department consists of multiple sections including hive uncapping, hive testing, honey extraction, and honey bucketing. Each section performs its specific task and generates reports for management. The processed honey is stored in the form of honey batches.

The honey batches are then stored in warehouses managed by employees. The packaging process is carried out by employees who convert honey batches into different types and sizes of packages. After packaging, the products are again stored in warehouses.

The organization also deals with distributors. Employees negotiate and finalize deals with distributors, after which the distributor makes an online payment. The employee sends a notification along with the transaction details to the head or management. Finally, the packaged honey is delivered to the distributor by the manager.

The system also includes resource management, where alerts generated from reports may request resources such as hives, plants, or other materials. These requests are processed and allocated to the respective departments.

The database system ensures proper storage, retrieval, and management of all data including employees, departments, reports, alerts, resources, honey batches, packaging, transactions, and distributor dealings. It also supports management in decision-making by generating reports and tracking all operations efficiently.

2. Business Requirements

<u>1. FARM MANAGEMENT</u>

The system must store information about farms including name, location, and area.

Each farm must manage natural resources such as trees and water usage.

Trees must be tracked with plantation date, height, and water consumption.

<u>2. HIVE & HONEY PRODUCTION</u>

The system must maintain records of bee hives including:

Number of worker bees

Brood count

Honey production

Each hive must be placed on a specific tree and assigned to an employee.

The system must track honey extracted from hives and convert it into honey batches.

<u>3. HONEY PROCESSING & TESTING</u>

Each honey batch must be tested before further processing.

The system must generate test reports for each batch including:

Purity level

Category (e.g., expensive, normal, cheap)

Test reports must be used to classify honey for packaging and distribution.

<u>4. STORAGE MANAGEMENT</u>

Honey batches must be stored in warehouses.

Each warehouse must have a responsible manager (employee).

The system must track:

Quantity of honey stored

Storage location

Honey batches must also be assigned to a bucket for handling and storage control.

5. PACKAGING SYSTEM

The system must allow employees to package honey batches into different package types.

Each package must have:

Size

Type

Price

Packaging must track:

Which employee performed the packaging

Which batch was used

Quantity packaged

6. DISTRIBUTION SYSTEM

Each package must be assigned to a distributor.

A distributor can receive multiple packages.

The system must maintain distributor details including:

Name

Contact information

CNIC

7. HUMAN RESOURCE MANAGEMENT

The system must manage employee data including:

Personal details

Role

Department

Section

Employees must be assigned to:

Farms

Departments

Sections

Each employee may have a manager (hierarchical structure).

Employees must be involved in:

Hive placement

Packaging

Warehouse management

8. CONTACT MANAGEMENT

Employees may have multiple contact numbers.

Contact information must be stored separately to support multi-valued attributes.

9. RESOURCE MANAGEMENT

The system must manage different types of resources required by departments.

Departments must be able to request resources.

Each request may include multiple resources.

10. SYSTEM INTEGRATION REQUIREMENT

The system must integrate:

Farm operations

HR management

Production and packaging

Distribution

All modules must be connected through shared entities like:

Employee

Department

Honey Batch

11. DATA INTEGRITY REQUIREMENTS

Each entity must have a unique identifier (Primary Key).

Relationships must be enforced using Foreign Keys.

Data must satisfy:

Domain constraints (valid ranges, formats)

Entity integrity (no null primary keys)

Referential integrity (valid relationships)

12. SCALABILITY & FLEXIBILITY

The system must support:

Multiple farms

Multiple warehouses

Multiple distributors

The system must allow expansion for:

New resource types

Additional packaging formats

More departments

3. Detailed Analysis using Nouns and Verbs

Entities · Relationships · Attributes

1. Nouns (Entities / Objects)

Human Entities

Entity
employee
manager
candidate
distributor

Organizational Structure

Entity
department
section
role
farm

Documentation / Control

Entity
report

Entity
alert
notification
test_report

HR System

Entity
vacancy
application
assignment
contact_info

Finance

Entity
salary_record
transaction

Farm System

Entity
tree

Entity
hive
honey_batch

Storage System

Entity
warehouse
storage
bucket

Packaging System

Entity
package
packaging

Resource System

Entity
resource
resource_request
request_detail

2. Verbs (Relationships + Operations)

Employee Relations

Verb (Relationship)	Meaning	Type
employee works in department	assignment	FK relationship
employee belongs to section	membership	FK
employee reports to employee	hierarchy	self FK
employee performs packaging	action	M:N via table
employee generates report	creation	FK
employee manages warehouse	control	1:M

Department Relations

Verb (Relationship)	Meaning	Type
department contains sections	hierarchy	1:M
department receives reports	flow	1:M
department requests resources	request	1:M
department creates vacancy	HR trigger	1:M

Reporting Flow

Verb (Relationship)	Meaning	Type
report triggers alert	event flow	1:M

Verb (Relationship)	Meaning	Type
alert generates notification	propagation	1:M

HR Flow

Verb (Relationship)	Meaning	Type
candidate applies for vacancy	M:N	junction table
vacancy receives applications	inverse M:N	junction
candidate becomes employee	conversion	entity transformation
assignment assigns employee role	mapping	junction

Farm & Hive

Verb (Relationship)	Meaning	Type
hive placed on tree	M:N	junction
hive produces honey_batch	1:M	FK
bucket contains honey_batch	1:M	FK

Storage & Packaging

Verb (Relationship)	Meaning	Type
honey_batch stored in warehouse	M:N	junction
employee packages honey_batch	M:N	junction

Verb (Relationship)	Meaning	Type
package assigned to distributor	1:M	FK

Resource System

Verb (Relationship)	Meaning	Type
department requests resource	1:M	FK
request contains resource	M:N	junction
resource allocated to department	M:N	junction

3. Adjectives (Attributes + Data Types)

Employee Attributes

Attribute	Data Type
employee_name	VARCHAR(100)
employee_email	VARCHAR(100)
employee_phone	VARCHAR(20)
employee_hire_date	DATE

Department Attributes

Attribute	Data Type
department_name	VARCHAR(100)

Attribute	Data Type
department_location	VARCHAR(100)

Section Attributes

Attribute	Data Type
section_name	VARCHAR(100)

Role Attributes

Attribute	Data Type
role_name	VARCHAR(50)

Hive Attributes

Attribute	Data Type
worker_bees	INT
brood	INT
honey_quantity	DECIMAL(10,2)

Honey Batch Attributes

Attribute	Data Type
quantity	DECIMAL(10,2)

Attribute	Data Type
received_date	DATE

Package Attributes

Attribute	Data Type
size	VARCHAR(20)
type	VARCHAR(50)
price	DECIMAL(10,2)

Warehouse Attributes

Attribute	Data Type
location	VARCHAR(100)

Report Attributes

Attribute	Data Type
report_type	VARCHAR(50)
report_content	TEXT
report_date	DATE

Alert Attributes

Attribute	Data Type
alert_type	VARCHAR(50)
alert_description	TEXT
alert_date	DATE

Notification Attributes

Attribute	Data Type
notification_type	VARCHAR(20)
message	TEXT
notify_date	DATE

Salary Attributes

Attribute	Data Type
basic_salary	DECIMAL(10,2)
bonus	DECIMAL(10,2)
deduction	DECIMAL(10,2)
net_salary	DECIMAL(10,2)

Candidate Attributes

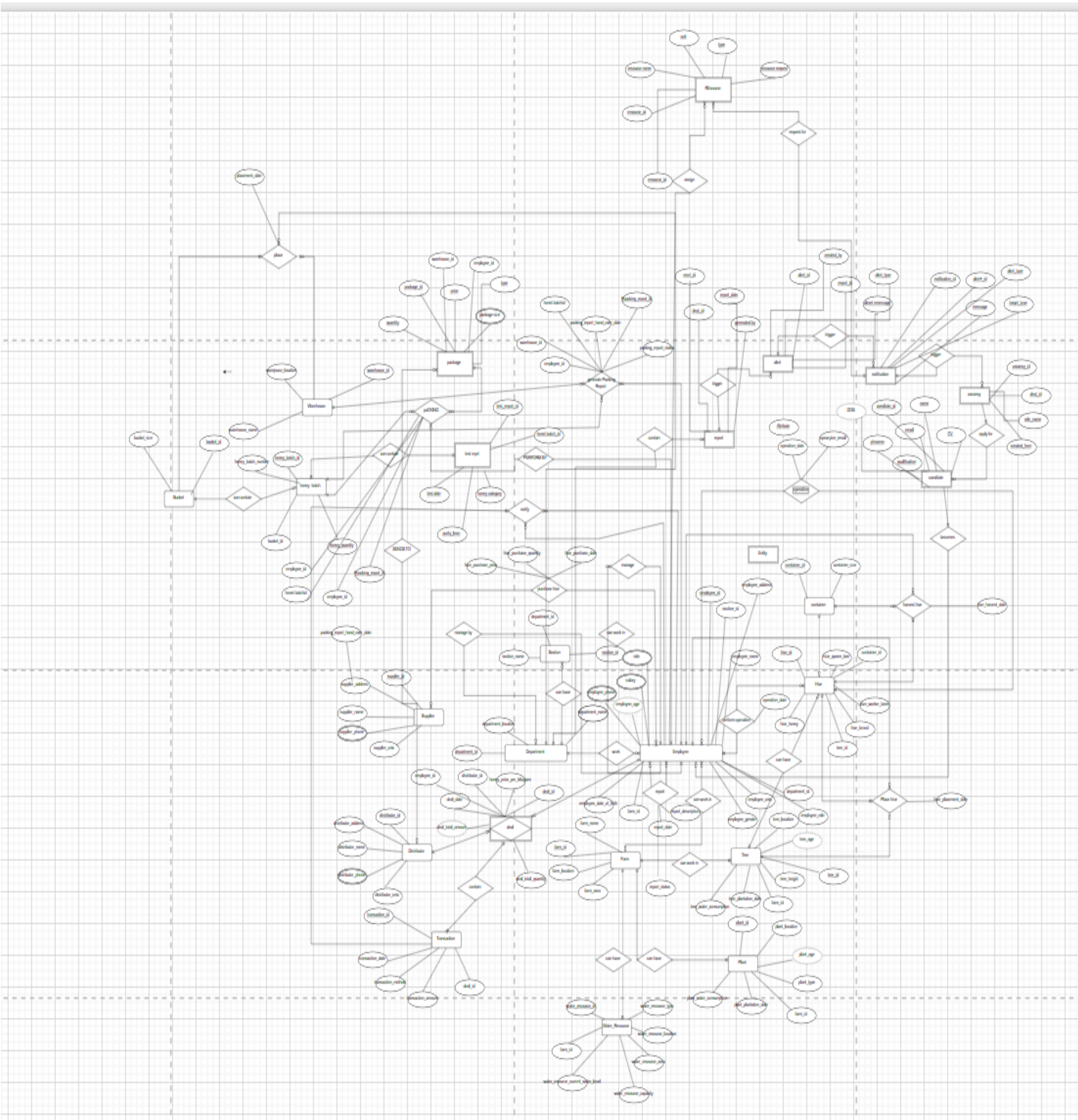
Attribute	Data Type
candidate_name	VARCHAR(100)
candidate_email	VARCHAR(100)
qualification	TEXT
experience	INT

Resource Attributes

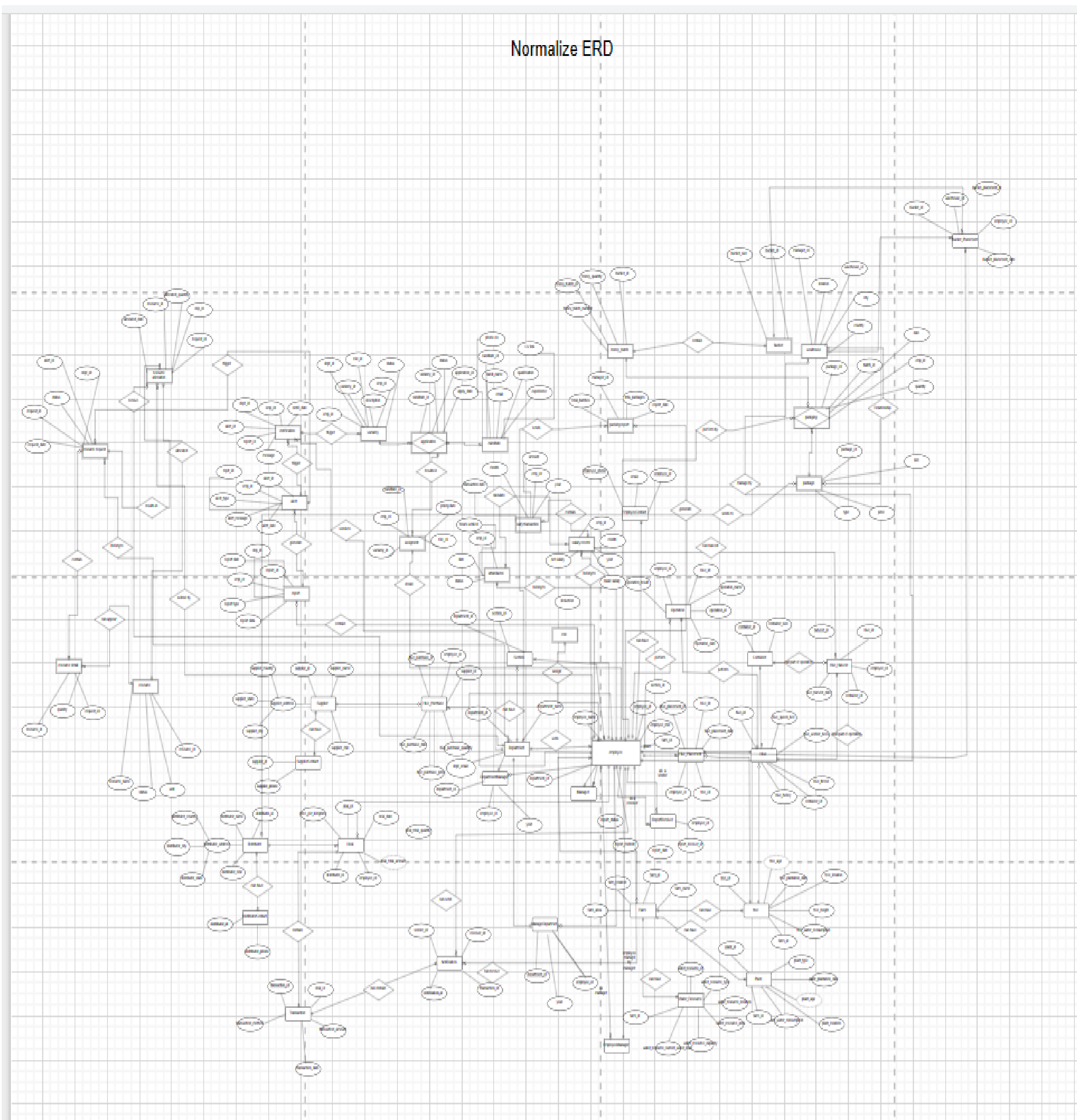
Attribute	Data Type
resource_name	VARCHAR(100)

3. Entity Relationship Diagram (ERD)

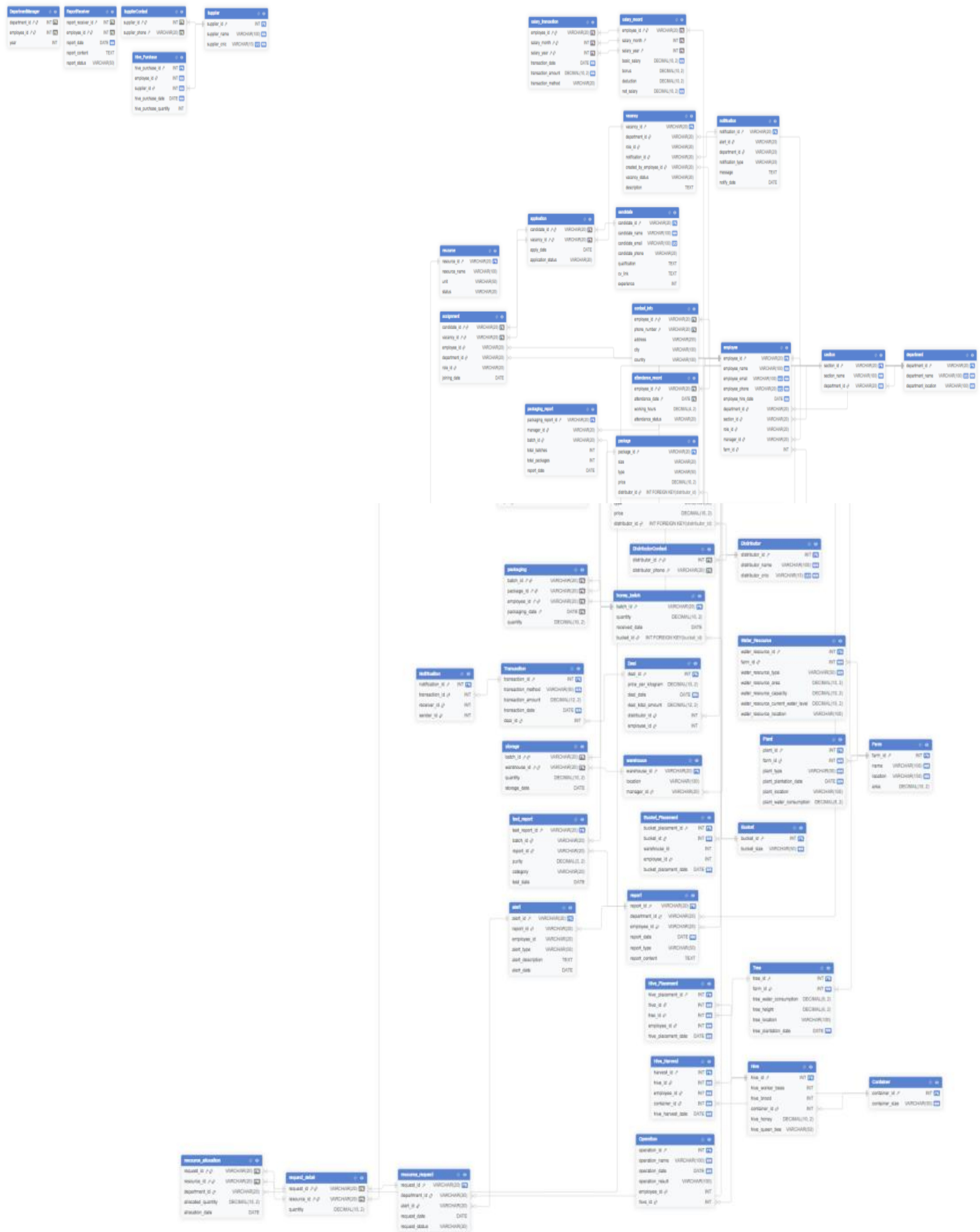
Unnormalize ERD



4. Normalized Entity Relationship Diagram



5. Diagram View



5. Relational Model

1.role

Field / Column	Data Type	Constraint
role_id	VARCHAR(20)	PK
role_name	VARCHAR(50)	NOT NULL, UNIQUE
role_description	TEXT	—

department

Field / Column	Data Type	Constraint
department_id	VARCHAR(20)	PK
department_name	VARCHAR(100)	NOT NULL, UNIQUE
department_location	VARCHAR(100)	NOT NULL
manager_id	VARCHAR(20)	FK → employee

3.section

Field / Column	Data Type	Constraint
section_id	VARCHAR(20)	PK
section_name	VARCHAR(100)	NOT NULL
department_id	VARCHAR(20)	FK → department
section_manager_id	VARCHAR(20)	FK → employee

4.employee

Field / Column	Data Type	Constraint
employee_id	VARCHAR(20)	PK
employee_name	VARCHAR(100)	NOT NULL
employee_email	VARCHAR(100)	NOT NULL, UNIQUE

Field / Column	Data Type	Constraint
employee_phone	VARCHAR(20)	NOT NULL, UNIQUE
employee_hire_date	DATE	NOT NULL
department_id	VARCHAR(20)	FK → department
section_id	VARCHAR(20)	FK → section
role_id	VARCHAR(20)	FK → role
manager_id	VARCHAR(20)	FK → employee (self)
farm_id	INT	FK → farm

5.contact_info

Field / Column	Data Type	Constraint
employee_id	VARCHAR(20)	PK, FK → employee
phone_number	VARCHAR(20)	PK
address	VARCHAR(255)	—
city	VARCHAR(100)	—
country	VARCHAR(100)	—

6.salary_record

Field / Column	Data Type	Constraint
employee_id	VARCHAR(20)	PK, FK → employee
salary_month	INT	PK, CHECK (1–12)
salary_year	INT	PK, CHECK (≥ 2000)
basic_salary	DECIMAL(10,2)	NOT NULL, CHECK ≥ 0
bonus	DECIMAL(10,2)	DEFAULT 0, CHECK ≥ 0
deduction	DECIMAL(10,2)	DEFAULT 0, CHECK ≥ 0

Field / Column	Data Type	Constraint
net_salary	DECIMAL(10,2)	NOT NULL, CHECK ≥ 0

7. salary_transaction

Field / Column	Data Type	Constraint
employee_id	VARCHAR(20)	PK, FK → salary_record
salary_month	INT	PK
salary_year	INT	PK
transaction_date	DATE	NOT NULL
transaction_amount	DECIMAL(10,2)	NOT NULL, CHECK ≥ 0
transaction_method	VARCHAR(20)	CHECK IN (Cash, Bank, Online)

attendance_record

Field / Column	Data Type	Constraint
employee_id	VARCHAR(20)	PK, FK → employee
attendance_date	DATE	PK
working_hours	DECIMAL(4,2)	CHECK ≥ 0
attendance_status	VARCHAR(20)	CHECK IN (Present, Absent, Leave)

report

Field / Column	Data Type	Constraint
report_id	VARCHAR(20)	PK
department_id	VARCHAR(20)	FK → department
employee_id	VARCHAR(20)	FK → employee
report_date	DATE	NOT NULL
report_type	VARCHAR(50)	—

Field / Column	Data Type	Constraint
report_content	TEXT	—

alert

Field / Column	Data Type	Constraint
alert_id	VARCHAR(20)	PK
report_id	VARCHAR(20)	FK → report
employee_id	VARCHAR(20)	—
alert_type	VARCHAR(50)	—
alert_description	TEXT	—
alert_date	DATE	—

notification

Field / Column	Data Type	Constraint
notification_id	VARCHAR(20)	PK
alert_id	VARCHAR(20)	FK → alert
department_id	VARCHAR(20)	FK → department
notification_type	VARCHAR(20)	CHECK IN (HR, RESOURCE)
message	TEXT	—
notify_date	DATE	—

4.1 vacancy

Field / Column	Data Type	Constraint
vacancy_id	VARCHAR(20)	PK
department_id	VARCHAR(20)	FK → department
role_id	VARCHAR(20)	FK → role

Field / Column	Data Type	Constraint
notification_id	VARCHAR(20)	FK → notification
created_by_employee_id	VARCHAR(20)	FK → employee
vacancy_status	VARCHAR(20)	CHECK IN (Open, Closed)
description	TEXT	—

4.2 candidate

Field / Column	Data Type	Constraint
candidate_id	VARCHAR(20)	PK
candidate_name	VARCHAR(100)	NOT NULL
candidate_email	VARCHAR(100)	UNIQUE
candidate_phone	VARCHAR(20)	—
qualification	TEXT	—
cv_link	TEXT	—
experience	INT	CHECK ≥ 0

4.3 application

Field / Column	Data Type	Constraint
candidate_id	VARCHAR(20)	PK, FK → candidate
vacancy_id	VARCHAR(20)	PK, FK → vacancy
apply_date	DATE	—
application_status	VARCHAR(20)	—

4.4 assignment

Field / Column	Data Type	Constraint
candidate_id	VARCHAR(20)	PK, FK → application

Field / Column	Data Type	Constraint
vacancy_id	VARCHAR(20)	PK, FK → application
employee_id	VARCHAR(20)	FK → employee
department_id	VARCHAR(20)	FK → department
role_id	VARCHAR(20)	FK → role
joining_date	DATE	—

5.1 resource

Field / Column	Data Type	Constraint
resource_id	VARCHAR(20)	PK
resource_name	VARCHAR(100)	—
unit	VARCHAR(50)	—
status	VARCHAR(20)	—

5.2 resource_request

Field / Column	Data Type	Constraint
request_id	VARCHAR(20)	PK
department_id	VARCHAR(20)	FK → department
alert_id	VARCHAR(20)	FK → alert
request_date	DATE	—
request_status	VARCHAR(20)	—

5.3 request_detail

Field / Column	Data Type	Constraint
request_id	VARCHAR(20)	PK, FK → resource_request
resource_id	VARCHAR(20)	PK, FK → resource

Field / Column	Data Type	Constraint
quantity	DECIMAL(10,2)	CHECK > 0

5.4 resource_allocation

Field / Column	Data Type	Constraint
request_id	VARCHAR(20)	PK, FK → request_detail
resource_id	VARCHAR(20)	PK, FK → request_detail
department_id	VARCHAR(20)	FK → department
allocated_quantity	DECIMAL(10,2)	CHECK > 0
allocation_date	DATE	—

6.1 farm

Field / Column	Data Type	Constraint
farm_id	INT	PK
name	VARCHAR(100)	NOT NULL
location	VARCHAR(150)	NOT NULL
area	DECIMAL(10,2)	CHECK > 0

6.2 tree

Field / Column	Data Type	Constraint
tree_id	INT	PK
farm_id	INT	NOT NULL, FK → farm
tree_water_consumption	DECIMAL(8,2)	CHECK ≥ 0
tree_height	DECIMAL(6,2)	CHECK > 0
tree_location	VARCHAR(100)	—
tree_plantation_date	DATE	NOT NULL

6.3 plant

Field / Column	Data Type	Constraint
plant_id	INT	PK
farm_id	INT	NOT NULL, FK → farm
plant_type	VARCHAR(50)	NOT NULL
plant_plantation_date	DATE	NOT NULL
plant_location	VARCHAR(100)	—
plant_water_consumption	DECIMAL(8,2)	CHECK ≥ 0

6.4 water_resource

Field / Column	Data Type	Constraint
water_resource_id	INT	PK
farm_id	INT	NOT NULL, FK → farm
water_resource_type	VARCHAR(50)	NOT NULL
water_resource_area	DECIMAL(10,2)	—
water_resource_capacity	DECIMAL(10,2)	CHECK ≥ 0
water_resource_current_water_level	DECIMAL(10,2)	CHECK ≥ 0
water_resource_location	VARCHAR(100)	—

6.5 container

Field / Column	Data Type	Constraint
container_id	INT	PK
container_size	VARCHAR(50)	NOT NULL

6.6 hive

Field / Column	Data Type	Constraint
hive_id	INT	PK
hive_worker_bees	INT	CHECK ≥ 0
hive_brood	INT	CHECK ≥ 0
container_id	INT	FK → container
hive_honey	DECIMAL(10,2)	CHECK ≥ 0
hive_queen_bee	VARCHAR(50)	—

6.7 hive_placement

Field / Column	Data Type	Constraint
hive_placement_id	INT	PK
hive_id	INT	NOT NULL, FK → hive
tree_id	INT	NOT NULL, FK → tree
employee_id	INT	NOT NULL, FK → employee
hive_placement_date	DATE	NOT NULL

6.8 hive_harvest

Field / Column	Data Type	Constraint
harvest_id	INT	PK
hive_id	INT	NOT NULL, FK → hive
employee_id	INT	NOT NULL, FK → employee
container_id	INT	NOT NULL, FK → container
hive_harvest_date	DATE	NOT NULL

6.9 operation

Field / Column	Data Type	Constraint
operation_id	INT	PK
operation_name	VARCHAR(100)	NOT NULL
operation_date	DATE	NOT NULL
operation_result	VARCHAR(100)	—
employee_id	INT	FK → employee
hive_id	INT	FK → hive

6.10 hive_purchase

Field / Column	Data Type	Constraint
hive_purchase_id	INT	PK
employee_id	INT	NOT NULL, FK → employee
supplier_id	INT	NOT NULL, FK → supplier
hive_purchase_date	DATE	NOT NULL
hive_purchase_quantity	INT	CHECK > 0

7.1 supplier

Field / Column	Data Type	Constraint
supplier_id	INT	PK
supplier_name	VARCHAR(100)	NOT NULL
supplier_cnic	VARCHAR(15)	UNIQUE, NOT NULL

7.2 supplier_contact

Field / Column	Data Type	Constraint
supplier_id	INT	PK, FK → supplier

Field / Column	Data Type	Constraint
supplier_phone	VARCHAR(20)	PK

8.1 distributor

Field / Column	Data Type	Constraint
distributor_id	INT	PK
distributor_name	VARCHAR(100)	NOT NULL
distributor_cnic	VARCHAR(15)	UNIQUE, NOT NULL

8.2 distributor_contact

Field / Column	Data Type	Constraint
distributor_id	INT	PK, FK → distributor
distributor_phone	VARCHAR(20)	PK

8.3 deal

Field / Column	Data Type	Constraint
deal_id	INT	PK
price_per_kilogram	DECIMAL(10,2)	CHECK > 0
deal_date	DATE	NOT NULL
deal_total_amount	DECIMAL(12,2)	CHECK ≥ 0
distributor_id	INT	FK → distributor
employee_id	INT	FK → employee

8.4 transaction

Field / Column	Data Type	Constraint
transaction_id	INT	PK
transaction_method	VARCHAR(50)	NOT NULL

Field / Column	Data Type	Constraint
transaction_amount	DECIMAL(12,2)	CHECK ≥ 0
transaction_date	DATE	NOT NULL
deal_id	INT	FK → deal

9.1 bucket

Field / Column	Data Type	Constraint
bucket_id	INT	PK
bucket_size	VARCHAR(50)	NOT NULL

9.2 honey_batch

Field / Column	Data Type	Constraint
batch_id	VARCHAR(20)	PK
quantity	DECIMAL(10,2)	—
received_date	DATE	—
bucket_id	INT	FK → bucket

9.3 test_report

Field / Column	Data Type	Constraint
test_report_id	VARCHAR(20)	PK
batch_id	VARCHAR(20)	FK → honey_batch
report_id	VARCHAR(20)	FK → report
purity	DECIMAL(5,2)	—
category	VARCHAR(20)	—
test_date	DATE	—

9.4 warehouse

Field / Column	Data Type	Constraint
warehouse_id	VARCHAR(20)	PK
location	VARCHAR(100)	—
manager_id	VARCHAR(20)	FK → employee

9.5 storage

Field / Column	Data Type	Constraint
batch_id	VARCHAR(20)	PK, FK → honey_batch
warehouse_id	VARCHAR(20)	PK, FK → warehouse
quantity	DECIMAL(10,2)	—
storage_date	DATE	—

9.6 bucket_placement

Field / Column	Data Type	Constraint
bucket_placement_id	INT	PK
bucket_id	INT	NOT NULL, FK → bucket
warehouse_id	INT	—
employee_id	INT	FK → employee
bucket_placement_date	DATE	NOT NULL

9.7 package

Field / Column	Data Type	Constraint
package_id	VARCHAR(20)	PK
size	VARCHAR(20)	—
type	VARCHAR(50)	—

Field / Column	Data Type	Constraint
price	DECIMAL(10,2)	—
distributor_id	INT	FK → distributor

9.8 packaging

Field / Column	Data Type	Constraint
batch_id	VARCHAR(20)	PK, FK → honey_batch
package_id	VARCHAR(20)	PK, FK → package
employee_id	VARCHAR(20)	PK, FK → employee
packaging_date	DATE	PK
quantity	DECIMAL(10,2)	—

9.9 packaging_report

Field / Column	Data Type	Constraint
packaging_report_id	VARCHAR(20)	PK
manager_id	VARCHAR(20)	FK → employee
batch_id	VARCHAR(20)	FK → honey_batch
total_batches	INT	—
total_packages	INT	—
report_date	DATE	—

10.1 report_receiver

Field / Column	Data Type	Constraint
report_receiver_id	INT	PK
employee_id	INT	PK, FK → employee
report_date	DATE	NOT NULL

Field / Column	Data Type	Constraint
report_content	TEXT	—
report_status	VARCHAR(50)	—

10.2 department_manager

Field / Column	Data Type	Constraint
department_id	INT	PK, FK → department
employee_id	INT	PK, FK → employee
year	INT	CHECK ≥ 2000

6. 20 SQL Queries

Query 1

SELECT * FROM employee;

Query 2

SELECT employee_id, employee_name, employee_email FROM employee;

Query 3

SELECT * FROM department;

Query 4

SELECT * FROM Farm WHERE area > 150;

Query 5

SELECT candidate_name, qualification, experience FROM candidate WHERE experience >= 3

Query 6

SELECT employee_name, employee_hire_date FROM employee

WHERE employee_hire_date > '2022-01-01';

Query 7

SELECT batch_id, quantity FROM honey_batch WHERE quantity BETWEEN 180 AND 230;

Query 8

SELECT candidate_name, candidate_email FROM candidate

WHERE candidate_email LIKE '%@gmail.com';

Query 9

SELECT * FROM alert WHERE alert_type IN ('Quality', 'Low Stock');

Query 10

SELECT package_id, size, price FROM package WHERE price < 10.00;

Query 11

SELECT employee_name, employee_hire_date FROM employee

ORDER BY employee_hire_date DESC;

Query 12

SELECT batch_id, purity FROM test_report ORDER BY purity DESC;

Query 13

SELECT TOP 5 * FROM deal ORDER BY deal_total_amount DESC;

Query 14

SELECT COUNT(*) AS TotalEmployees FROM employee;

Query 15

SELECT AVG(price_per_kilogram) AS AvgPrice FROM deal;

Query 16

SELECT department_id, COUNT(*) AS EmployeeCount

FROM employee GROUP BY department_id;

Query 17

SELECT SUM(quantity) AS TotalHoneyProduced FROM honey_batch;

Query 18

SELECT AVG(purity) AS AvgPurity, MAX(purity) AS MaxPurity, MIN(purity) AS MinPurity

Query 19

SELECT department_id, COUNT(*) AS EmpCount

FROM employee

GROUP BY department_id

HAVING COUNT(*) > 10;

Query 20

SELECT employee_id, SUM(working_hours) AS TotalHours

FROM attendance_record

GROUP BY employee_id

HAVING SUM(working_hours) > 40;

7. 20 Reporting Queries

Query 21

Question: Find distributors who have made more than 2 deals (transactions). Show distributor ID and number of deals.

SELECT distributor_id, COUNT(*) AS DealCount

FROM deal

GROUP BY distributor_id

HAVING COUNT(*) >= 2;

Query 22

Question: List departments that have more than 10 employees. Show department ID and employee count.

SELECT department_id, COUNT(*) AS EmployeeCount

FROM employee

WHERE department_id IS NOT NULL

GROUP BY department_id

HAVING COUNT(*) > 10;

Query 23

Question: Show honey batches that have been packaged into more than 3 different package types. Display batch ID and number of package types.

SELECT batch_id, COUNT(DISTINCT package_id) AS PackageTypeCount

FROM packaging

GROUP BY batch_id

HAVING COUNT(DISTINCT package_id) > 3;

Query 24

Question: Display all employees with their department names and role names.

```
SELECT e.employee_name, d.department_name, r.role_name

FROM employee e

LEFT JOIN department d ON e.department_id = d.department_id

LEFT JOIN role r ON e.role_id = r.role_id;
```

Query 25

Question: Show honey batch information along with bucket size used for storage.

```
SELECT h.batch_id, h.quantity, h.received_date, b.bucket_size

FROM honey_batch h

INNER JOIN Bucket b ON h.bucket_id = b.bucket_id;
```

Query 26

Question: List all candidates along with the vacancies they applied for and application status.

```
SELECT c.candidate_name, v.vacancy_id, a.application_status, a.apply_date

FROM candidate c

INNER JOIN application a ON c.candidate_id = a.candidate_id

INNER JOIN vacancy v ON a.vacancy_id = v.vacancy_id;
```

Query 27

Question: Display all packages with their distributor names (including packages without distributors).

```
SELECT p.package_id, p.size, p.type, p.price, d.distributor_name

FROM package p

LEFT JOIN Distributor d ON p.distributor_id = d.distributor_id;
```

Query 28

Question: Show attendance records with employee names and department names for January 2024.

```
SELECT e.employee_name, d.department_name, a.attendance_date, a.attendance_status, a.working_hours

FROM attendance_record a

INNER JOIN employee e ON a.employee_id = e.employee_id

INNER JOIN department d ON e.department_id = d.department_id

WHERE a.attendance_date BETWEEN '2024-01-01' AND '2024-01-31';
```

Query 29

Question: List all deals with distributor names and the employee who handled the deal.

```
SELECT dl.deal_id, dl.deal_date, dl.price_per_kilogram, dl.deal_total_amount,

    d.distributor_name, e.employee_name AS HandledBy

FROM deal dl

INNER JOIN Distributor d ON dl.distributor_id = d.distributor_id

INNER JOIN employee e ON dl.employee_id = e.employee_id;
```

Query 30

Question: Display complete employee information including department, section, role, and manager name.

```
SELECT e.employee_name AS Employee,

    d.department_name AS Department,

    s.section_name AS Section,

    r.role_name AS Role,

    m.employee_name AS Manager

FROM employee e

LEFT JOIN department d ON e.department_id = d.department_id

LEFT JOIN section s ON e.section_id = s.section_id

LEFT JOIN role r ON e.role_id = r.role_id

LEFT JOIN employee m ON e.manager_id = m.employee_id;
```

Query 31

Question: Show test reports with batch details, report type, and department that generated the report.

```
SELECT t.test_report_id, h.batch_id, t.purity, t.category,
       r.report_type, d.department_name
FROM test_report t
INNER JOIN honey_batch h ON t.batch_id = h.batch_id
INNER JOIN report r ON t.report_id = r.report_id
INNER JOIN department d ON r.department_id = d.department_id;
```

Query 32

Question: Display resource requests with department name, alert details, and allocated quantities.

```
SELECT rr.request_id, d.department_name, a.alert_type,
       a.alert_description, ra.allocated_quantity, ra.allocation_date
FROM resource_request rr
LEFT JOIN department d ON rr.department_id = d.department_id
LEFT JOIN alert a ON rr.alert_id = a.alert_id
LEFT JOIN resource_allocation ra ON rr.request_id = ra.request_id;
```

Query 33

Question: Find all employees who work in the same department as 'John Smith'.

```
SELECT employee_name, department_id
FROM employee
WHERE department_id = (
   SELECT department_id
   FROM employee
   WHERE employee_name = 'John Smith'
```

);

Query 34

Question: List honey batches that have quantity greater than the average batch quantity.

```
SELECT batch_id, quantity, received_date
FROM honey_batch
WHERE quantity > (SELECT AVG(quantity) FROM honey_batch)
ORDER BY quantity DESC;
```

Query 35

Question: Show all candidates who applied for vacancies in the Sales Department.

```
SELECT candidate_name, qualification, experience
FROM candidate
WHERE candidate_id IN (
   SELECT candidate_id
   FROM application
   WHERE vacancy_id IN (
      SELECT vacancy_id
      FROM vacancy
      WHERE department_id = (
         SELECT department_id
         FROM department
         WHERE department_name = 'Sales Department'
      )
   )
);
```

Query 36

Question: Find employees who earn more than the average salary in their department.

```
SELECT e.employee_name, e.department_id, sr.basic_salary
FROM employee e
INNER JOIN salary_record sr ON e.employee_id = sr.employee_id
WHERE sr.basic_salary > (
    SELECT AVG(sr2.basic_salary)
    FROM salary_record sr2
    INNER JOIN employee e2 ON sr2.employee_id = e2.employee_id
    WHERE e2.department_id = e.department_id
    AND sr2.salary_year = 2024
)
AND sr.salary_year = 2024;
```

Query 37

Question: Find all employees who are managers and are currently managing a department.

```
SELECT employee_name, employee_id
FROM employee
WHERE employee_id IN (
    SELECT manager_id
    FROM department
    WHERE manager_id IN (
        SELECT employee_id
        FROM employee
        WHERE role_id IN ('R001', 'R002')
    )
);
```

Query 38

Question: List all honey batches that have been tested with purity above 98% and stored in warehouses located in North District.

```
SELECT batch_id, quantity, received_date

FROM honey_batch

WHERE batch_id IN (

    SELECT batch_id

    FROM test_report

    WHERE purity > 98

)

AND batch_id IN (

    SELECT batch_id

    FROM storage

    WHERE warehouse_id IN (

        SELECT warehouse_id

        FROM warehouse

        WHERE location LIKE '%North%'

    )

);
```

Query 39

Question: Generate monthly salary summary report showing total salary expenses by department for 2024.

```
SELECT

    d.department_name,

    sr.salary_month,

    COUNT(e.employee_id) AS EmployeeCount,
```

```sql
    SUM(sr.basic_salary) AS TotalBasicSalary,
    SUM(sr.bonus) AS TotalBonus,
    SUM(sr.deduction) AS TotalDeduction,
    SUM(sr.net_salary) AS TotalNetSalary
FROM department d
INNER JOIN employee e ON d.department_id = e.department_id
INNER JOIN salary_record sr ON e.employee_id = sr.employee_id
WHERE sr.salary_year = 2024
GROUP BY d.department_name, sr.salary_month
ORDER BY d.department_name, sr.salary_month;
```

Query 40

Question: Create a comprehensive performance report showing department-wise statistics including employee count, average salary, total deals, total honey production, and average honey purity.

```sql
WITH DepartmentMetrics AS (
    SELECT
        d.department_id,
        d.department_name,
        COUNT(DISTINCT e.employee_id) AS TotalEmployees,
        AVG(sr.basic_salary) AS AvgSalary,
        COUNT(DISTINCT dl.deal_id) AS TotalDeals
    FROM department d
    LEFT JOIN employee e ON d.department_id = e.department_id
    LEFT JOIN salary_record sr ON e.employee_id = sr.employee_id AND sr.salary_year = 2024
    LEFT JOIN deal dl ON dl.employee_id = e.employee_id
    GROUP BY d.department_id, d.department_name
```

```
),
HoneyMetrics AS (
    SELECT
        d.department_id,
        SUM(hb.quantity) AS TotalHoneyProduction,
        AVG(tr.purity) AS AvgPurity
    FROM department d
    LEFT JOIN employee e ON d.department_id = e.department_id
    LEFT JOIN honey_batch hb ON e.farm_id = hb.bucket_id
    LEFT JOIN test_report tr ON hb.batch_id = tr.batch_id
    GROUP BY d.department_id
)
SELECT
    dm.department_name,
    dm.TotalEmployees,
    ROUND(dm.AvgSalary, 2) AS AverageSalary,
    ISNULL(hm.TotalHoneyProduction, 0) AS TotalHoneyProduced_KG,
    ROUND(ISNULL(hm.AvgPurity, 0), 2) AS AveragePurity_Percentage,
    ISNULL(dm.TotalDeals, 0) AS TotalDealsCompleted,
    CASE
        WHEN dm.TotalEmployees > 15 THEN 'Large Department'
        WHEN dm.TotalEmployees > 8 THEN 'Medium Department'
        ELSE 'Small Department'
    END AS DepartmentSize,
    CASE
```

```
        WHEN ISNULL(hm.AvgPurity, 0) >= 98 THEN 'Premium Quality'
        WHEN ISNULL(hm.AvgPurity, 0) >= 95 THEN 'Standard Quality'
        ELSE 'Needs Improvement'
    END AS QualityRating
FROM DepartmentMetrics dm
LEFT JOIN HoneyMetrics hm ON dm.department_id = hm.department_id
ORDER BY dm.TotalEmployees DESC, hm.AvgPurity DESC;
```